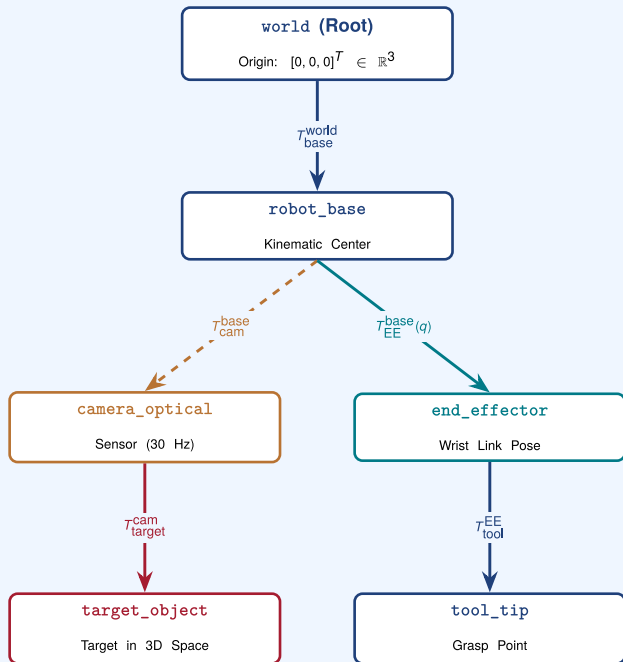


Coordinate Frame Graph ($T_B^A \in SE(3)$)



Time-Indexed Buffer & Lie Tangent Propagation

1. Delayed Vision Ingestion ($\Delta t_{lag} = 65$ ms):

Vision token arrives at $t_{now} = 100$ ms, but photons were transduced at $t_{capture} = 35$ ms.

Query buffer for historical transform: $T_{cam}^{base}(t = 35 \text{ ms})$.

2. Forward Coordinate Chaining ($SE(3)$):

$$T_{target}^{world}(t_{cap}) = T_{base}^{world} \cdot T_{cam}^{base}(t_{cap}) \cdot T_{target}^{cam}(t_{cap})$$

Homogeneous matrix multiplication guarantees geometric consistency.

3. Tangent-Space Propagation ($\mathfrak{se}(3)$):

Forward-integrate from $t_{capture}$ to t_{now} using IMU/joint twists $\xi = [v, \omega]^T \in \mathbb{R}^6$:

$$T(t_{now}) = T(t_{capture}) \cdot \exp(\xi^\wedge \cdot \Delta t_{lag})$$

Exponential map via Rodrigues formula prevents gimbal lock singularities.

4. Real-Time Query Guarantee:

Lookups execute in $< 10 \mu s$ via lock-free ring buffer on Linux MPU and FreeRTOS MCU, enabling 1000 Hz collision checks.

THE SE(3) FRAME CONTRACT: SPATIAL PERSISTENCE UNDER DELAY

Dynamic transform graphs decouple sensor arrival rates from motor control cadences. Even if vision lags by 65 ms, the system reconstructs the exact metric target pose in real time without coordinate drift.